CA Final

STRATEGIC COST & PERFORMANCE MANAGEMENT

FORMULA SHEET

SPOM Set B & IBS

Prepared by

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AIR 50

Strategic Cost & Performance Management

CA Final | SPOM Set B & IBS | Formula Sheet

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Note: Space is left after each chapter for writing— summaries, formulas, and key points in your own words.

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MODERN BUSINESS ENVIRONMENT

Cost of Quality (COQ) = Cost of Good Quality + Cost of Poor Quality

Cost of Good Quality = Prevention cost + Appraisal Cost

Cost of Poor Quality = Internal failure cost + External failure cost

Optimal COQ level is where sum of cost of good quality and cost of poor quality is minimum.

$$\text{Units purchased} = \frac{\text{Units required}}{(100 - \text{defective rate})\%}$$

Analysis in questions we have to compare current cost with proposed cost, if

Current cost > proposed cost, then **apply the strategy**
Current cost < proposed cost, then **do not apply the strategy**

Above evaluation is on the basis of monetary factors, we also have to write about non-monetary factors i.e. on the basis of non-monetary factor we can either apply or not apply it.

LEAN SYSTEM AND INNOVATION

Inventory Holding Cost = Average Inventory x Inventory Holding Cost p.u.

Average Inventory = $\frac{\text{Opening Inventory + Closing Inventory}}{2}$

Overtime Cost= $\frac{\text{Shortage}}{\text{Efficiency}}$ x (Normal labour cost + premium on overtime)

Labour efficiency ratio = $\frac{\text{Standard hours to complete work}}{\text{Actual hours to complete work}}$ X 100

(Kitne ghante ka kaam kitne ghante mein kiya)

Overall Equipment Effectiveness (OEE) = Availability Ratio x Performance Ratio x Quality Ratio

Availability Ratio = $\frac{\text{Planned Production Time - Unplanned Downtime}}{\text{Planned Production Time}}$ X 100

OR

$\frac{\text{Operating Time}}{\text{Planned Production Time}}$ x 100

(Kitna ghante main kaam karne wala tha less kitne ghante mere waste hogye kisi reason ke wajah se jo reason socha nhi tha for ex. Light chali gyi, machine kharab hogyi etc.)

If in question it is given total hours worked then it is operating time, to arrive at Planned Production = Operating time + unplanned downtime.

Planned Production Time = Total hours available - Planned downtime

(itne ghante ke liye maine labour ko hire kiya tha usme se kuch time mein unka lunch break and tea break (ye planned hota h pehle hi) minus kardiya toh bacha ki actually who kaam karne kitne ghanta wale the)

Performance Ratio = $\frac{\text{Standard time to be taken}}{\text{Actual time taken}}$ X 100

OR

$$\frac{\text{Ideal Cycle Time x Total Count}}{\text{Operating Time}} \times 100$$

(Kitne ghante mein kaam hona chahiye tha vs kitne mein hua)

Quality Ratio = $\frac{\text{Total Units produced – Scrap units}}{\text{Total units produced}}$

OR

$\frac{\text{Good Count}}{\text{Total Count}} \times 100$

(Jitni units banayi usme se kitni sahi nikli ye mera ho gya kitni achi quality ka kaam main karta hu)

OEE can also be calculated = (Good Count × Ideal Cycle Time) / Planned Production Time

Whether to opt Just in time (JIT) or not:

Calculate total cost in both the options and then compare

If cost in JIT > Cost in normal practice, then not to opt JIT
cost in JIT < Cost in normal practice, then opt JIT.

Takt Time = Used to match the pace of production to the pace of customer demand.

Takt Time= $\frac{\text{Available Production Time}}{\text{Total Quantity Required}}$

Here, Available Production Time is 'total available time for production' – 'planned downtime.

SPECIALIST COST MANAGEMENT TECHNIQUES

Cost Structure for calculation of target cost:

Target Selling price	XXX
Less: Royalty (if any)	XXX
Less: Profit	XXX
Target Cost	**XXX**

Cost Structure:

Raw Material	XXX
Add: Labour Charges	XXX
Add: Direct Expenses	XXX
Add: Product Specific Overheads	XXX
Add: Other Directly attributable cost	XXX
Total Cost of Manufacturing	**XXX**

Compare total cost with Target cost if

Total cost > Target cost, then reduce the cost
Total cost < Target cost, then it's a favorable case no need to do anything.

Calculation of 'Total Labour Hours' over the Lifetime of the Product

When there is no learning curve:

No of product to be produced in the lifetime X Labour hour per product

When there is learning curve:

Learning Curve means with time labour will be able to produce more product with the same given time. Due to this per unit labour hour will be different.

average time per unit for x units is:

$$\mathbf{Yx} = ax^{b}$$

Where,
Y = Average time per unit
x = no of units for which calculation is made

a = hours to produce 1st unit
b = index of learning curve

For ex: if it is given that learning curve is till 250 units and time taken to produce per unit beyond 250 units is equal to time taken to produce 250th unit then we have to calculate time taken to produce 250th unit. For this we will be calculating total time taken for producing 250 units (total) less total time taken to produce 249 units (total) the difference shall the time taken to produce 250th unit. *(refer illustration 2 on page no 4.19).*

Bottleneck activity = Activity means a resource whose demand is more than its availability.

Determined by = $\frac{\text{Required hours}}{\text{Available hours}} \times 100$

Among the resource whichever resource is having highest utilization percent will be considered as key factor.

Throughput Accounting is return per product per bottleneck hour.

Throughput Accounting Ratio = $\frac{\text{Throughput per Bottleneck Minute}}{\text{Factory Cost per Bottleneck Minute}}$

Here,

Throughput per Bottleneck Minute = $\frac{\text{Selling Price – Material and Other Variable Cost}}{\text{Time}}$

Bottleneck minute = minutes of a particular item which are in shortage or are limited
Factory cost = cost to run factory *(generally given)*

In question it can be asked product return per minute and throughput accounting ratio both are different. Calculation for both the items will be done separately.

Additional formulas:

= $\frac{\text{Throughput}}{\text{Labour cost}}$

= $\frac{\text{Throughput}}{\text{Material Cost}}$

Product return per minute = $\frac{\text{Contribution per unit}}{\text{Minutes required per unit}}$

Throughput Accounting Ratio and product per minute is higher the better. If ranking is to be done then 1st rank will be given to item having highest figure.

If in question it is given, we have already accepted some orders then I have to complete that order and after that remaining bottleneck resource will be used by applying the throughput concept.

Throughput lost:

Throughput I could earn if I produce product having highest throughput per unit	XXX
Less: Throughput I have earned by producing product having lower throughput per unit	XXX
Throughput lost	**XXX**

Environmental Cost *(it is similar to Activity based costing)* allocate each environmental cost to each product on the basis of a particular driver. For ex total packing cost is given then allocate such cost on the basis of packing material etc.

Life cycle cost is total cost incurred during the entire lifecycle of the product which includes R&D cost, variable costs, fixed cost, marketing and distribution cost etc.

STRATEGIC REVENUE MANAGEMENT

CVP ANALYSIS

Contribution = Sales per unit – Variable cost per unit

Break even point (BEP) (in units) = $\frac{\textbf{Fixed Cost}}{\textbf{Sales per unit – Variable cost per unit}}$

(at what number of units I will be able to cover my fixed cost is called BEP)

Batch-level activities:

Total set-up costs per annum = Number of setups per annum × cost of each set-up

CVP analysis using Activity Based Costing:

Break-even point (units) = [fixed cost + (batch level cost driver × cost per batch driver) + (product sustaining cost driver × cost per product sustaining cost driver)] / (selling price per unit – variable cost per unit)

Statement Showing Cost Benefit Analysis

Incremental cost of project	xxx
Less: Savings in future cost	xxx
Less: Incremental revenue	xxx
Incremental Benefit	**xxx**

Make or Buy (Outsourcing decision)

Compare cost incurred to make the product inhouse vs cost to purchase it from outsider considering the opportunity cost.

If incremental cost savings + opportunity costs < incremental costs, reject the outsourcing, unless qualitative factors fiercely impact the decision.

If incremental cost savings + opportunity costs > incremental costs, accept the outsourcing unless qualitative factors fiercely impact the decision.

If incremental cost savings + opportunity costs are = incremental costs, focus primarily on qualitative factors to evaluate the decision.

Sell or Process Further

Whether we should sell the product on a given stage or process it further and then sell it. To decide either to sell a component/ product/ raw material as it is or alternatively process it further by incurring additional expenses usually in the case of joint products. Only the incremental costs and revenues of the further process are relevant. The joint process costs are irrelevant - they are already 'sunk' at the point of separation.

Minimum Pricing Decisions

Minimum price = Incremental costs of manufacturing + opportunity costs (if any).

Incremental cost of manufacturing only includes labour hours which are not in excess, cost of excess hours available is not considered.

Opportunity cost shall be negative if I have to pay additional amount to otherwise dispose the material it shall be therefore deducted to arrive at minimum price.

Keep or Drop Decisions for unprofitable segments

Saving of avoidable fixed cost	XXX
Less: increase in variable cost	XXX
Less: loss of contribution	XXX
Overall Profit / (Loss)	**XXX**

(Hum ek division ki avoidable fixed cost bacha lenge pr raw material pe jo hume discount milra tha who band ho jayega humari cost ban jayegi woh plus jo division unprofitable h who jo contribution deri thi thoda bohot who bhi band ho jayega toh who bhi cost h uske hisab se decision making karni padegi)

Special Order Decisions

Decision has to be made whether the special order is going to add positive contribution for the company. Decision has to be made on the basis of limiting factor. For ex. Machine time is a limiting factor then the contribution per machine hour from both activities has to be calculated and on the basis of that decision has to be made.

Product Mix Decision

If there is a shortage or limited supply of certain other resources, which may act as a key factor, like, for example, machine hours, then the contribution is linked to such a key factor for taking a decision. For example, in an undertaking, the availability of machine capacity is limited, and the machine hours required for one unit of the two products are different. In such cases, the contribution is to be linked with the machine hour, and the product that yields the highest contribution per machine hour is to be preferred for making decision.

Contribution per limiting factor = $\frac{\text{Contribution per unit}}{\text{Limiting factor required to produce one unit}}$

If there is more than one limiting factor then we have to rank the products on the basis of different limiting factor separately and rank them on the basis on such factor and then use linear programming to find the optimum mix. *(refer illustration 9 on page no. 6.30)*

Profit Maximisation Model

i. **As per the economic theory of pricing:**

Profit is Maximum at a level of output where Marginal Revenue (MR) is equal to Marginal Cost (MC), i.e., Marginal Revenue (MR) = Marginal Cost (MC)

ii. The Basic Price equation, which is used to determine the Price where Profit is Maximum. The equation is written as: P = a – bQ

Where, P = Price

b = Slope of the Demand Curve, Calculated as [$b = \frac{\text{change in price}}{\text{change in quantity}}$]

Q = Quantity Demanded

a = Price at Which Demand is Zero

Marginal Revenue equation is written as Marginal Revenue (MR) = P = a – 2bQ

Target Rate of return

Price = Unit Cost + [(Desired Rate of Return * Invested Capital)/Number of unit sales]

Objective Value or True Economic Value (TEV)

TEV = Cost of the Next Best Alternative + Value of Performance Differential

Where,

Cost of the next best alternative = cost of a comparable product offered by another company.

Value of the performance differential = value of additional features provided by the seller of a product.

Perceived Value
Value that the consumer understands the product delivers to it. It is the price of a product that a consumer is willing to spend to have that product.

STRATEGIC PROFIT MANAGEMENT

Revenue Effect

Revenue Effect of Growth = (Actual units of output sold in current year – Actual units of output sold in last year) X Selling price in last year

The revenue effect of growth measures the increase/ decrease in revenue solely due to change in number of units sold.

Cost Effect

Cost effect of growth measures the effect of variable cost and fixed cost separately.

Cost Effect of Growth for Variable Costs =

(Unit of input required to produce current year output in last year - Actual units of input used to produce last year's output) X Input price in last year

The cost effect of growth for variable cost measures the increase/ decrease in variable cost solely due to change in number of units produced.

Cost Effect of Growth for *Fixed Costs*

= [(Actual units of capacity in last year if adequate to produce current year's production in last year) − (Actual units of capacity in last year)] × Price per unit of capacity in last year

The cost effect of growth for fixed cost measures the increase/ decrease in fixed cost solely due to change in number of units produced.

Price Recovery Component

Revenue Effect

Revenue Effect of Price Recovery

[(Selling Price in current year) – (Selling Price in last year)] X Actual units of output sold in current year

The revenue effect of price recovery measures the increase/ decrease in revenue solely due to change in selling prices.

Cost Effect

Cost Effect of Price Recovery for *Variable Cost* =

[(Input price in current year) − (Input price in last year)] × Units of input required to Produce current year's output in last year

The cost effect of price recovery measures the increase/ decrease in variable cost solely due to change in prices.

Cost Effect of Price Recovery for *Fixed Costs*

= [(Price per unit of capacity in current year) − (Price per unit of capacity in last year)] × Actual units of capacity in last year, if adequate to produce current year's output in last year

The cost effect of price recovery measures the increase/ decrease in fixed cost solely due to change in prices.

Productivity Component

Productivity Component measures the change in the operating income due to changes in the product mix and/ or yield of inputs as compared with the last year. This component uses the current year's prices of input to measure the changes in costs only.

(ye measure karta h meri ebit ya operating income mein kitna change aya due to maine change kardiya product mix (quantity of two products) ya yield of input as compared to last year).

Cost Effect of Productivity for Variable Cost

= [(Actual units of input used to produce current year's output) − (Units of input required to produce current year's output in last year)] × Input price in current year

(Final output banana ke liye pichle saal kitna input lagta or iss saal kitna input laga jo bhi difference ayega usko current year price se multiply kardiya toh mere pass ayega maine kitni additional cost lagai h iss saal inputs pe)

Cost Effect of Productivity for Fixed Costs

= [(Actual units of capacity in current year) − (Actual units of capacity in last year if adequate to produce current year's production in last year)] × Price per unit of capacity in current year

(Maine kitni additional cost lagayi h fixed cost pe because maine ho sakta h apni capacity increase kari ho jisse meri fixed cost increase hogyi ho.)

These formulas are very logical and link with their name. Don't cram these formula because you will get confused in exam, understand the logic and flow of formula few compare cost few compare production units etc. and you will be able to solve the questions.

4. Reconciliation of Operating Profit

Particulars	**Costs**	**Revenue**
Operating Profit in Last Year	**xxx**	
***Add/(Less):* Revenue and Cost Effect of Growth Component**	**(+) F / (-) A**	**(+) F / (-) A**
***Add/(Less):* Revenue and Cost Effect of Price Recovery Component**	**(+) F / (-) A**	**(+) F / (-) A**
***Add/ (Less):* Cost Effect of Productivity Component**	**(+) F / (-) A**	**-**
Total	**xxx**	**xxx**
Operating Profit in Current Year (Revenue – Costs)	**xxx**	

F= Favorable, A= Adverse

Manufacturing Cycle Efficiency (MCE) = The term "Efficiency" can be interpreted to mean "manufacturing related Value Addition". So, MCE is the ratio of value addition time to the total time taken to manufacture a product.

MCE= $\frac{\text{Processing Time}}{\text{Manufacturing Cycle Time (include all type of time including processing)}}$

Delivery Cycle Time = *Time taken **the moment the order is received** through to final delivery*

Manufacturing Cycle Time = total time taken to manufacture a product

A Perfect MCE would yield a ratio equal to 1 i.e., all processing time would be value-added. While such a result is not likely to occur, managers should always strive to raise the ratio in that direction.

Any non-value-added time results in an MCE of less than 1. For example an MCE of 0.5, would mean that half of the total production time consists of inspection, moving, and similar non-value-added activities.

Activity Based costing / Activity based budgeting

Cost of each activity should be allocated between different products on the basis of certain cost driver.

For ex: Power charges are allocated on the basis of KW/h used by each product
Rent is allocated on the basis of Sq ft area occupied for each product.

Pareto Analysis (80:20 rule)

80% of result comes from 20% efforts while the remaining 20 percent of results are achieved by 80 percent of effort.

Direct Product Profitability

Sales	XXX
Less: Cost of Goods Sold	XXX
Gross Margin	XXX
Less: Direct Product Costs (Warehouse, Transportation, Store etc.)	XXX
Direct Product Profit	**XXX**

AN INTRODUCTION TO STRATEGIC PERFORMANCE MANAGEMENT

Altman Z score

For Publicly Held Manufacturing Firms

Z Score = $1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$

Wherein, X1 = working capital/total assets
X2 = retained earnings/total assets
X3 = earnings before interest and tax/total assets
X4 = market value of equity/total liabilities
X5 = sales/total assets
1.2, 1.4, 3.3, 0.6, and 1.0 are pre-defined weights (coefficients).

Z-Score	Zone of discrimination	Prediction regarding corporate failure (due to bankruptcy)
Less than 1.81	Distress	Companies with a Z score of below 1.81 are in danger and possibly heading towards bankruptcy.
1.81 to 2.99	Grey	Companies with scores 1.81 to 2.99 need further investigation.
More than 2.99	Safe	Companies with a score more than 2.99 are financially sound.

Private Firms, Non-manufacturers, and Emerging Markets

Z score =$0.717X_1 + 0.847X_2 + 3.107X_3 + 0.420X_4 + 0.998X_5$

Wherein, X1 = working capital/total assets
X2 = retained earnings/total assets
X3 = earnings before interest and tax/total assets
X4 = book value of equity/total liabilities
X5 = sales/total assets

Z-Score	Zone of discrimination	Prediction regarding corporate failure (due to bankruptcy)
Less than 1.23	Distress	Companies with a Z score of below 1.23 are in danger and possibly heading towards bankruptcy.

Z-Score	Zone of discrimination	Prediction regarding corporate failure (due to bankruptcy)
1.23 to 2.99	Grey	Companies with scores 1.23 to 2.99 need further investigation.
More than 2.99	Safe	Companies with a score more than 2.99 are financially sound.

Non-manufacturing firm operating in developed markets

Z score for Non-Manufacturer = $6.56X_1 + 3.26X_2 + 6.72X_3 + 1.05X_4$

Wherein, X1 = working capital / total assets
X2 = retained earnings / total assets
X3 = earnings before interest and taxes / total assets
X4 = book value of equity / total liabilities

Z-Score	Zone of discrimination	Prediction regarding corporate failure (due to bankruptcy)
Less than 1.1	Distress	Companies with a Z score of below 1.1 are in danger and possibly heading towards bankruptcy.
1.1 to 2.6	Grey	Companies with scores 1.81 to 2.6 need further investigation.
More than 2.6	Safe	Companies with a score more than 2.6 are financially sound.

Non-manufacturing firm operating in emerging markets
Z score = $3.25 + 6.56X_1 + 3.26X_2 + 6.72X_3 + 1.05X_4$

Wherein, X1 = working capital / total assets
X2 = retained earnings / total assets
X3 = earnings before interest and taxes / total assets
X4 = book value of equity / total liabilities

Analysis same as above table (Non-manufacturing firm operating in developed markets)

Performance Analysis Score (PAS or Tafflers' PAS)

$Z = 3.2 + 12.18X_1 + 2.50X_2 – 10.68X_3 + 0.029X_4$

Wherein,
X1 = profit before tax/current assets
X2 = current assets/current liabilities
X3 = current liabilities/total assets
X4 = no credit interval

The negative Z-score means that the company is facing a potential bankruptcy.

Argenti's A score (read table from module pg 8.40)

STRATEGIC PERFORMANCE MEASURES IN PRIVATE SECTOR

Gross Profit (GP) = Net Sale – Cost of Goods Sold
where Cost of Goods Sold = Opening Inventory + Purchases + Wages + Direct Expenses – Closing Inventory.

GP Margin = (Gross Profit / Sales) × 100
Return on Capital Employed (ROCE) = (Profit Before Interest and Tax (PBIT) / Average Capital Employed) × 100

Capital employed = shareholders' capital plus long-term debt finance or sum of total assets less current liabilities

Average capital employed = $\frac{\text{opening capital employed + closing capital employed}}{2}$

PBIT = Operating Profit

Return on Investment (ROI) = Net Profit/Investment

Earning per share = $\frac{\text{Profit attributable to ordinary shareholders}}{\text{Weighted average number of ordinary shares outstanding during the period}}$

Economic Value Added (EVA) = NOPAT – WACC × Capital

Where **NOPAT**

EBIT	XXX
+/- adjustments	XXX
Adjusted EBIT	XXX
Less: Interest	XXX
EBT	XXX
Less: Tax	XXX
EAT	XXX
Add: Interest (1-t)	XXX
NOPAT	**XXX**

OR

EBIT	XXX
+/- adjustments	XXX
Adjusted EBIT	XXX
Less: Tax	XXX
NOPAT	**XXX**

Adjustments:

1. Non-cash items like goodwill written off, provision for doubtful debts etc. should be added back (if deducted while calculating EBIT)
2. Charge economic depreciation i.e. add back depreciation charged in EBIT and deduct economic depreciation.

3. Deferred revenue expenditure like advertisement, research etc. should be added back.

4. Only Tax Paid is to be considered i.e. add back any provision for tax if any. (It doesn't include deferred tax).

Net Present Value (NPV) = PV of Cash Inflow – PV of Cash Outflow

Residual income = Controllable Profit - Cost of Capital

Price-Earnings (P/E) Ratio = $\frac{\text{Current Market Price}}{\text{Earnings Per Share}}$

Profitability Index (PI) = $\frac{\text{Present Value of Cash Inflows}}{\text{Present Value of Cash Outflows}}$

Sensitivity Analysis (%) = $\frac{\text{NPV}}{\text{PV of Variable under consideration}} \times 100$

DIVISIONAL TRANSFER PRICING

This chapter has a lot of practical questions but all these questions are based on logics instead of learning the formula, focus on building the concept. Few of the key points to be kept in mind in order to solve the questions are:

While calculating transfer price for supplying division

1. How much marginal cost i am incurring to produce one additional unit
   (Mera kitna kharcha hora h ek unit banane mein)

2. I want minimum this amount of profit
   (Mereko kam se kam itna profit toh chahiye who chahe dusra department de chahe bahar ka buyer)

3. I am unable to sell all the units I have produced so buy these remaining product at a price equivalent to my cost
   (Dekho mera sara maal toh bik nhi raha toh tum mereko bas meri cost dedo i.e. fixed cost waise bhi nhi bikra waise bhi nhi bikra kuch toh paise aye)

While calculating transfer price for receiving division

1. I am able to buy this product at INR X from 3rd party I can give you a maximum of X only.
   (Mereko bahar se itne ruppe ka padra h main tereko bhi itne hi de paunga)

2. I am going to earn INR 50 on this product I can sacrifice the entire profit in order to purchase this from you.
   (mereko 50 ruppe bachne h ispe main maximum uss price pe kharid lunga jaha mereko 50 bhi na bache par nuksan nhi hona chahiye)

Few of the basic formulas and concepts of this chapters are:

Range of transfer price that promotes goal congruence

(Goal congruence means puri company as a whole a fayada hona chahiye koi ek department jada profit kamane ke liye dusre department ka loss na kardein jisse eventually puri company jo profit kamayegi who less hoga as compared to what it could earn)

i. **Minimum Transfer Price** (determined by the supplying division) = Additional Outlay Cost per unit + Opportunity Cost per unit.

(bekne wala department kehra h mereko kam se kam itna toh chahiye = Jitni additional cost main dusre department ki demand puri karne ke liye lagaunga + uski demand ko pura na karta or bahar bekta toh itne toh kama hi leta i.e. opportunity cost)

Additional Outlay Cost = Marginal Cost + Any Additional Incidental Costs incurred by the supplying division e.g., storage, transportation etc. Marginal cost is cost to produce one additional unit)

Opportunity Cost = benefit that is foregone from selling internally rather than externally.

ii. **Maximum Transfer Price** (determined by the purchasing division) = Lower of Net Marginal Revenue and the External Buy-in Price

Net Marginal Revenue = Marginal Revenue (i.e. Selling Price p.u.) – Marginal Cost to Purchasing
Division

Marginal revenue means iss saman ko bek ke main itna kama leta hu par dusre department se main itne ruppe mein kharid lunga ki meri na koi kamayi ho na koi nuksan

***External Buy price** means price charged by outsider/ 3rd party. (bahar se main itna ka le leta hu)*

I can make formula for each and every concept for this chapter but it is useless just to burden yourself to learn each and every formula just practice the questions and if you need made your own formula for a particular concept that will be much impactful.

Strategic Performance Measures in the Non-for-Profit Organisations

Average Employee Cost per week per house = $\frac{\text{Total Employee Cost}}{\text{Number of Houses} \times \text{52 weeks}}$

Percentage of Rent Lost = $\frac{\text{Rent Lost due to Vacancy}}{\text{Gross Rent Receivable}}$

Value for Money (VFM) Framework = The qualitative assessment of **Economy, Efficiency, and Effectiveness.**

STANDARD COSTING

This chapter contains a lot of formulas and remembering each and every formula is very difficult. In order to resolve this issue, try to conceptual understand the topic and each formula it won't take a lot of time. This chapter weightage is very high in both SPOM and IBS. Also, after conceptually understanding the formulas, you can also tweak the formulas to get the answer for MCQs because these are not the rigid formulas.

1. Planning & Operational Variances

Planning Variance compares a Revised Standard to the Original Standard (also called Revision Variance). Whereas Operational Variance compares the Actual Results against the Revised Standard.

1.1 Direct Material Usage Variance

Variance Type	Formula
Traditional (Total) Usage Variance	(Standard Quantity - Actual Quantity) × Standard Price
Planning Usage Variance	(Standard Quantity - Revised Standard Quantity) × Standard Price
Operational Usage Variance	(Revised Standard Quantity - Actual Quantity) × Revised Standard Price

1.2 Direct Material Price Variance

Variance Type	Formula
Traditional (Total) Price Variance	(Standard Price - Actual Price) × Actual Quantity
Planning Price Variance	(Standard Price - Revised Standard Price) × Revised Standard Quantity
Operational Price Variance	(Revised Standard Price - Actual Price) × Actual Quantity

1.3 Direct Labour Efficiency Variance

Variance Type	Formula
Traditional (Total) Efficiency Variance	(Standard Time - Actual Time) × Standard Rate
Planning Efficiency Variance	(Standard Time - Revised Standard Time) × Standard Rate
Operational Efficiency Variance	(Revised Standard Time - Actual Time) × Revised Standard Rate

1.4 Direct Labour Rate Variance

Variance Type	Formula
Traditional (Total) Rate Variance	(Standard Rate - Actual Rate) × Actual Time
Planning Rate Variance	(Standard Rate - Revised Standard Rate) × Revised Standard Time
Operational Rate Variance	(Revised Standard Rate - Actual Rate) × Actual Time

2. Variance Analysis in Activity Based Costing (ABC)

Variance	Formula
Efficiency Variance	(Standard Activity Units for Actual Output - Actual Activity Units) × Standard Rate per Activity
Expenditure Variance	(Actual Activity Units × Standard Rate) - Actual Cost

3. Sales Variances — Absorption Costing (in terms of Profit / Margin)

3.1 Sales Margin Variance

Variance	Formula
Sales Margin Variance (Total)	(AQ × AM) - (BQ × SM)
Sales Margin Price Variance	(AM - SM) × AQ OR (AM × AQ) - (SM × AQ)
Sales Margin Volume Variance	(AQ - BQ) × SM OR (SM × AQ) - (SM × BQ)
Sales Margin Mix Variance	(AQ - RAQ) × SM OR (AQ × SM) - (RAQ × SM)
Sales Margin Quantity Variance	(RAQ - BQ) × SM OR (RAQ × SM) - (BQ × SM) OR Avg Budgeted Margin per unit of Budgeted Mix × (Total Actual Qty - Total Budgeted Qty)
Market Size Variance	Budgeted Mkt Share % × (Actual Industry Qty - Budgeted Industry Qty) × Avg Budgeted Margin per unit
Market Share Variance	(Actual Mkt Share % - Budgeted Mkt Share %) × Actual Industry Qty × Avg Budgeted Margin per unit

Notation: AQ = Actual Qty, BQ = Budgeted Qty, RAQ = Actual Qty in Budgeted Proportion, SM = Standard Margin (Std Price - Std Cost), AM = Actual Margin (Actual Price - Std Cost)

4. Sales Variances — Marginal Costing (in terms of Contribution)

4.1 Sales Contribution Variance

Variance	Formula
Sales Contribution Variance (Total)	(AQ × AC) - (BQ × SC)
Sales Contribution Price Variance	(AC - SC) × AQ OR (AC × AQ) - (SC × AQ)
Sales Contribution Volume Variance	(AQ - BQ) × SC OR (SC × AQ) - (SC × BQ)
Sales Contribution Mix Variance	(AQ - RAQ) × SC OR (AQ × SC) - (RAQ × SC)
Sales Contribution Quantity Variance	(RAQ - BQ) × SC OR (RAQ × SC) - (BQ × SC)
Market Size Variance	Budgeted Mkt Share % × (Actual Industry Sales Qty in units - Budgeted Industry Sales Qty in units) × Avg Budgeted Contribution per unit
Market Share Variance	(Actual Mkt Share % - Budgeted Mkt Share %) × (Actual Industry Sales Qty) × (Avg Budgeted Contribution per unit)
Notation: SC = Standard Contribution (Std Price - Std Variable Cost), AC = Actual Contribution (Actual Price - Std Variable Cost), RAQ = Actual Qty in Budgeted Proportion	

5. Sales Variances — Turnover / Value

Variance	Formula
Sales Variance (Total)	(AQ × AP) - (BQ × SP)
Sales Price Variance	(AP - SP) × AQ OR (AP × AQ) - (SP × AQ)
Sales Volume Variance	(AQ - BQ) × SP OR (SP × AQ) - (SP × BQ)
Sales Mix Variance	(AQ - RAQ) × SP OR (SP × AQ) - (SP × RAQ)
Sales Quantity Variance	(RAQ - BQ) × SP OR (SP × RAQ) - (SP × BQ)
Market Size Variance	Budgeted Mkt Share % × (Actual Industry Sales Qty - Budgeted Industry Sales Qty) × Avg Budgeted Margin per unit
Market Share Variance	(Actual Mkt Share % - Budgeted Mkt Share %) × Actual Industry Sales Qty × Avg Budgeted Margin per unit
Notation: SP = Standard Selling Price per unit, AP = Actual Selling Price per unit, AQ = Actual Sales Qty, BQ = Budgeted Sales Qty, RAQ = Actual Qty in Budgeted Proportion	

6. Key Relationships — Sales Variances

Relationship	Formula / Rule
Sales Price Variance =	Sales Margin Price Variance = Sales Contribution Price Variance
Sales Margin Volume Variance	= Sales Volume Variance × Budgeted Net Profit Ratio
Sales Contribution Volume Variance	= Sales Volume Variance × Budgeted P/V Ratio
Sales Margin Volume Variance	= Sales Contribution Volume Variance - Fixed Overhead Volume Variance
Sales Contribution Volume Variance	= Sales Margin Volume Variance + Fixed Overhead Volume Variance
Market Size + Market Share Variances	= Sales Margin Quantity Variance (or Sales Contribution Quantity Variance)

7. Direct Material Variances

Variance	Formula
Direct Material Total Variance	(SQ × SP) - (AQ × AP)
Direct Material Price Variance	(SP - AP) × AQ OR (SP × AQ) - (AP × AQ)
Direct Material Usage Variance	(SQ - AQ) × SP OR (SQ × SP) - (AQ × SP)
Direct Material Mix Variance	(RAQ - AQ) × SP OR (RAQ × SP) - (AQ × SP)
Direct Material Yield Variance	(SQ - RAQ) × SP OR (SQ × SP) - (RAQ × SP)
Material Purchase Price Variance	(SP - AP) × PQ OR (SP × PQ) - (AP × PQ)

Notation: SQ = Standard Qty (Expected for Actual Output), AQ = Actual Qty Consumed, RAQ = Actual Qty in Standard Proportion, SP = Standard Price, AP = Actual Price, PQ = Purchase Quantity

8. Direct Labour Variances

Variance	Formula
Direct Labour Total Variance	(SH × SR) - (AH* × AR)
Direct Labour Rate Variance	(SR - AR) × AH* OR (SR × AH*) - (AR × AH*)
Direct Labour Idle Time Variance	(AH* - AH#) × SR OR (AH* × SR) - (AH# × SR)
Direct Labour Efficiency Variance	(SH - AH#) × SR OR (SH × SR) - (AH# × SR)
Direct Labour Mix (Gang) Variance	(RAH - AH#) × SR OR (RAH × SR) - (AH# × SR)

Variance	Formula
Direct Labour Yield (Sub-Efficiency) Variance	(SH - RAH) × SR OR (SH × SR) - (RAH × SR)

Notation: SH = Standard Hours (for Actual Output), AH = Actual Hours Paid, AH# = Actual Hours Worked, RAH = Actual Hours Worked in Standard Proportion, SR = Standard Rate, AR = Actual Rate. When no idle time: AH* = AH#*

9. Fixed Production Overhead Variances

Variance	Formula
Fixed Overhead Total Variance	Absorbed Fixed Overheads - Actual Fixed Overheads
Fixed Overhead Expenditure Variance	Budgeted Fixed Overheads - Actual Fixed Overheads
Fixed Overhead Volume Variance	Absorbed Fixed Overheads - Budgeted Fixed Overheads
Fixed Overhead Capacity Variance	(Budgeted Fixed OH for Actual Hours) - Budgeted Fixed OH
Fixed Overhead Efficiency Variance	Absorbed Fixed OH - (Budgeted Fixed OH for Actual Hours)
Fixed Overhead Calendar Variance	Possible Fixed OH - Budgeted Fixed OH
Fixed Overhead Volume Variance (I)	Std Fixed OH Rate per Unit × (Actual Output - Budgeted Output)
Fixed Overhead Volume Variance (II)	Std Fixed OH Rate per Hour × (Std Hours for Actual Output - Budgeted Hours)
Fixed Overhead Efficiency Variance (expanded)	Std Fixed OH Rate per Hour × (Std Hours for Actual Output - Actual Hours)
Fixed Overhead Capacity Variance (expanded)	Std Fixed OH Rate per Hour × (Actual Hours - Budgeted Hours)

Possible Fixed Overheads = (Budgeted Fixed OH / Budgeted Days) × Actual Days | Absorbed = Std Fixed OH Rate per Unit × Actual Production

10. Variable Production Overhead Variances

Variance	Formula
Variable Overhead Total Variance	Std Variable OH for Production - Actual Variable OH
Variable Overhead Expenditure (Spending) Variance	(Std Rate × Actual Hours) - Actual Variable OH OR Actual Hours × (Std Rate - Actual Rate)

Variance	Formula
Variable Overhead Efficiency Variance	Std Variable OH for Production - Budgeted Variable OH for Actual Hours
Std Variable OH for Production = Std Rate per Hour × Std Hours for Actual Output | Budgeted Var OH for Actual Hours = Std Rate per Hour × Actual Hours	

11. Reconciliation of Profit — Key Concepts

11.1 Standard Cost Definitions

Term	Formula / Definition
Standard Fixed OH Absorbed (per Unit)	Std Fixed OH Rate per Unit × Actual Production in Units
Standard Fixed OH Absorbed (per Hour)	Std Fixed OH Rate per Hour × Std Hours for Actual Production
Budgeted Fixed Overheads	Std Fixed OH Rate per Unit × Budgeted Production Units
Budgeted Fixed OH for Actual Hours	Std Fixed OH Rate per Hour × Actual Hours
Standard Contribution	Standard Selling Price per unit - Standard Variable Cost per unit
Standard Margin	Standard Selling Price per unit - Standard Cost (total) per unit
Budgeted Profit (Absorption Costing)	Budgeted Quantity × Standard Margin
Standard Profit (Absorption Costing)	Actual Quantity × Standard Margin

11.2 Marginal vs Absorption Costing — Overhead Treatment

System	Fixed Overhead Variances
Absorption Costing	Fixed Overhead Expenditure Variance + Fixed Overhead Volume Variance (= Capacity + Efficiency)
Marginal Costing	Fixed Overhead Expenditure Variance ONLY (no Volume Variance — fixed OH not absorbed)

12. Learning Curve — Impact on Variances

Element	Formula / Description
Learning Curve Model	y = a × x^b

Element	Formula / Description
y	Average time per unit for x units
a	Time required for the first unit
x	Cumulative number of units produced
b	Learning coefficient (negative value)
Labour Rate Variance (Learning)	Actual Hours × (Std Rate - Actual Rate)
Labour Efficiency Variance (Learning)	Std Rate × (Std Hours [per learning curve] - Actual Hours)
Standard hours are derived from the learning curve model for the actual batch/cumulative output produced.	

13. Alternative Formulas for Mix & Quantity Variances

Variance	Alternative Formula
Sales Margin Mix Variance (Alt)	Total Actual Qty × (Avg Std Margin per unit of Actual Mix - Avg Budgeted Margin per unit of Budgeted Mix)
Sales Contribution Mix Variance (Alt)	Total Actual Qty × (Avg Std Contribution per unit of Actual Mix - Avg Budgeted Contribution per unit of Budgeted Mix)
Sales Contribution Quantity Variance (Alt)	Avg Budgeted Contribution per unit × (Total Actual Qty - Total Budgeted Qty)
Direct Material Mix Variance (Alt)	Total Actual Qty × (Avg Std Price per unit of Std Mix - Avg Std Price per unit of Actual Mix)
Direct Material Yield Variance (Alt)	Avg Std Price per unit of Std Mix × (Total Std Qty - Total Actual Qty)
Direct Labour Mix (Gang) Variance (Alt)	Total Actual Time Worked × (Avg Std Rate per hour of Std Gang - Avg Std Rate per hour of Actual Gang)
Direct Labour Yield (Sub-Efficiency) Variance (Alt)	Avg Std Rate per hour of Std Gang × (Total Std Time - Total Actual Time Worked)

14. Abbreviations & Notation Key

Abbreviation	Meaning
AQ	Actual Sales / Material Quantity
BQ	Budgeted Sales / Material Quantity

Abbreviation	Meaning
RAQ	Revised Actual Quantity = Actual Quantity rewritten in Budgeted Proportion
SQ	Standard Quantity = Expected Consumption for Actual Output
SP / AP	Standard Price / Actual Price per unit
SM / AM	Standard Margin / Actual Margin per unit
SC / AC	Standard Contribution / Actual Contribution per unit
SH / AH* / AH#	Standard Hours / Actual Hours Paid / Actual Hours Worked
RAH	Revised Actual Hours = Actual Hours Worked in Standard Proportion
SR / AR	Standard Rate / Actual Rate per Labour Hour
PQ	Purchase Quantity (for Material Purchase Price Variance)
F / A	Favourable / Adverse
Ex-ante	Before the event (Original Standard set before period begins)
Ex-post	After the event (Revised Standard set after period, reflecting actual conditions)

----------*THANKYOU*----------